Dense subclasses of L^p

REAL VARIABLES I

Fall 2026

The spaces $L^p(X,\mu)$ are complete for every $p \in (0,+\infty]$ (Notes 25 and 26). A general element of $L^p(X,\mu)$, however, is only an equivalence class of measurable functions, and many statements are easier to verify for functions with additional structure: simple functions take finitely many values, and continuous functions can be evaluated at points and controlled on compact sets. A class of functions is *dense* in $L^p(X,\mu)$ if every element of $L^p(X,\mu)$ is an L^p limit of functions from that class. A typical density argument proves a statement for the functions of the class and transfers it to all of $L^p(X,\mu)$ by passing to the limit.

These notes establish two dense classes. We first recall why simple functions whose support has finite measure are dense in $L^p(X,\mu)$ for $p \in (0,+\infty)$, and we show that this density fails in L^∞. We then prove that bounded continuous functions are dense in $L^p(X,\mu)$ for every finite Borel measure μ on a metric space X, and we state that compactly supported continuous functions are dense in $L^p(\mathbb{R}^d,\mu)$ for every Borel measure μ on $\mathbb{R}^d$ that is finite on compact sets; the proof of this last statement is left to Homework 08. The proof reduces the approximation of a general function to the approximation of an indicator $\mathbb{1}_E$ of a Borel set. The regularity of finite Borel measures provides a closed set F and an open set U with $F \subseteq E \subseteq U$ and $\mu(U \setminus F)$ small, and a continuous function that equals 1 on F and 0 outside U then approximates $\mathbb{1}_E$.

After studying these notes, you will be able to do the following.

- Prove that simple functions with support of finite measure are dense in $L^p(X,\mu)$ for $p \in (0,+\infty)$, and show that they are not dense in $L^\infty(X,\mu)$ when $\mu(X) = +\infty$.
- State and prove the regularity of finite Borel measures on metric spaces by closed and open sets, and on $\mathbb{R}^d$ by compact sets.
- Construct continuous functions separating a closed set from the complement of an open set using distance functions.
- Prove that bounded continuous functions are dense in $L^p(X,\mu)$ for every finite Borel measure on a metric space and every $p \in (0,+\infty)$.
- State the density of compactly supported continuous functions in $L^p(\mathbb{R}^d,\mu)$ for Borel measures finite on compact sets, and explain why this density fails for some σ-finite Borel measures and for $p = \infty$.

Throughout these notes, $p \in (0,+\infty)$ unless $p = \infty$ is stated explicitly, and functions are measurable functions with values in $\mathbb{R}$. A subset $\mathcal{D} \subseteq L^p(X,\mu)$ is **dense** in $L^p(X,\mu)$ if for every $f \in L^p(X,\mu)$ and every $\varepsilon > 0$ there exists $g \in \mathcal{D}$ with $\|f-g\|_p < \varepsilon$.

14.1 Simple functions

A *simple function* is a measurable function $s\colon X \to \mathbb{R}$ that takes finitely many values. Writing $a_1, \dots, a_m$ for its distinct non-zero values and $A_i \coloneqq \{s = a_i\}$, we have the canonical representation $s = \sum_{i=1}^m a_i \mathbb{1}_{A_i}$ with pairwise disjoint measurable sets A_i, and

$$\int_X |s|^p \, \mathrm{d}\mu = \sum_{i=1}^m |a_i|^p \, \mu(A_i).$$

For $p < +\infty$, a simple function therefore belongs to $L^p(X,\mu)$ if and only if each A_i has finite measure, that is, if and only if $\mu(\{s \neq 0\}) < +\infty$. We denote by $\mathcal{S}_{\text{fin}}(X,\mu)$ the set of simple functions s with $\mu(\{s \neq 0\}) < +\infty$.

Theorem 14.1 (Density of simple functions in L^p). For $p \in (0,+\infty)$ and $f \in L^p(X,\mu)$, there exist $s_n \in \mathcal{S}_{\text{fin}}(X,\mu)$ such that $|s_n| \le |f|$ on X for every n, $s_n \to f$ pointwise on X, and $s_n \to f$ in L^p. In particular, $\mathcal{S}_{\text{fin}}(X,\mu)$ is dense in $L^p(X,\mu)$.

Proof of Theorem 14.1. Step 1: Approximation of the positive and negative parts.
By the approximation of non-negative measurable functions from below by simple functions (Notes 19–20), there exist non-negative simple functions $t_n \uparrow f^+$ and $u_n \uparrow f^-$ pointwise on X. We define $s_n \coloneqq t_n - u_n$. Then $s_n \to f^+ - f^- = f$ pointwise on X, and $|s_n| \le t_n + u_n \le f^+ + f^- = |f|$.

Step 2: Finite measure of the support.
If $s_n \neq 0$, let $c_n > 0$ denote the smallest absolute value of a non-zero value of s_n. Then $\{s_n \neq 0\} \subseteq \{|f| \ge c_n\}$, and Markov's inequality (Note 21) applied to $|f|^p$ gives

$$\mu(\{s_n \neq 0\}) \le \mu(\{|f|^p \ge c_n^p\}) \le c_n^{-p} \int_X |f|^p \, \mathrm{d}\mu < +\infty.$$

Hence $s_n \in \mathcal{S}_{\text{fin}}(X,\mu)$.

Step 3: Convergence in L^p.
The functions s_n converge to f pointwise and are dominated by $|f| \in L^p(X,\mu)$. Dominated convergence in L^p (Note 25) gives $s_n \to f$ in L^p. □

The approximating functions satisfy $\{s_n \neq 0\} \subseteq \{f \neq 0\}$, because $|s_n| \le |f|$. In particular, if f vanishes outside a set B, then every s_n vanishes outside B.

The exponent $p < +\infty$ enters through the requirement $\mu(\{s \neq 0\}) < +\infty$, which is forced by $\int_X |s|^p \, \mathrm{d}\mu < +\infty$. For $p = \infty$, every bounded simple function belongs to $L^\infty(X,\mu)$, and the class $\mathcal{S}_{\text{fin}}(X,\mu)$ is too small.

Proposition 14.2 (Simple functions with support of finite measure are not dense in L^∞). Suppose that $\mu(X) = +\infty$. Then the constant function $1 \in L^\infty(X,\mu)$ satisfies $\|1 - s\|_\infty \ge 1$ for every $s \in \mathcal{S}_{\text{fin}}(X,\mu)$. Consequently, $\mathcal{S}_{\text{fin}}(X,\mu)$ is not dense in $L^\infty(X,\mu)$.

Proof of Proposition 14.2. For $s \in \mathcal{S}_{\text{fin}}(X,\mu)$, the set $\{s = 0\} = X \setminus \{s \neq 0\}$ has measure $+\infty$, because $\mu(X) = +\infty$ and $\mu(\{s \neq 0\}) < +\infty$. On this set of positive measure, $|1 - s| = 1$. Hence no number $M < 1$ is an almost-everywhere bound for $|1 - s|$, and $\|1 - s\|_\infty \ge 1$. □

For example, on $\mathbb{R}$ with Lebesgue measure, no simple function vanishing outside a set of finite measure lies within L^∞ distance less than 1 of the constant function 1. If the restriction on the support is dropped, simple functions are dense in $L^\infty(X,\mu)$: for $f \in L^\infty(X,\mu)$ and $\varepsilon > 0$, the simple function $s \coloneqq \sum_{k\in\mathbb{Z}} k\varepsilon \, \mathbb{1}_{\{k\varepsilon \le f < (k+1)\varepsilon\}}$, which has finitely many non-zero terms because $|f| \le \|f\|_\infty$ almost everywhere, satisfies $\|f - s\|_\infty \le \varepsilon$. When $\mu(X) = +\infty$, such simple functions need not belong to any $L^p(X,\mu)$ with $p < +\infty$, so they cannot be used to approximate in L^p and in L^∞ simultaneously.

14.2 Continuous functions: statements

To speak of continuous functions, we need a topology on X and a measure defined on the open sets. Throughout this section and the next two, (X,d) is a metric space, $\mathcal{B}(X)$ is its Borel σ-algebra, that is, the σ-algebra

generated by the open subsets of X (Unit II), and a **finite Borel measure** on X is a measure μ on $\mathcal{B}(X)$ with $\mu(X) < +\infty$. Continuous functions $g\colon X \to \mathbb{R}$ are Borel measurable (Notes 17–18). We denote by $C_b(X)$ the bounded continuous functions on X. If μ is finite and $g \in C_b(X)$, then $\int_X |g|^p \, d\mu \le (\sup_X |g|)^p \mu(X) < +\infty$, so $C_b(X) \subseteq L^p(X, \mu)$.

We pursue two related goals. The first is the density statement.

Theorem 14.3 (Density of continuous functions for finite Borel measures). For a finite Borel measure μ on a metric space (X, d) and $p \in (0, +\infty)$, the set $C_b(X)$ is dense in $L^p(X, \mu)$.

By Theorem 14.1, it suffices to approximate simple functions, and hence indicators $\mathbb{1}_E$ of Borel sets. A continuous function cannot jump from 1 on E to 0 outside E, so it must make the transition on a region around the boundary of E. The second goal states that this region can be chosen to have small measure.

Theorem 14.4 (Regularity of finite Borel measures). For a finite Borel measure μ on a metric space (X, d), every Borel set $E \subseteq X$, and every $\varepsilon > 0$, there exist a closed set F and an open set U with

$$F \subseteq E \subseteq U \qquad \text{and} \qquad \mu(U \setminus F) < \varepsilon. \tag{14.1}$$

Consequently,

$$\mu(E) = \sup\{\mu(F) : F \subseteq E,\ F \text{ closed}\} = \inf\{\mu(U) : U \supseteq E,\ U \text{ open}\}.$$

If $X = \mathbb{R}^d$ with the Euclidean distance, the closed set F in (14.1) can be chosen compact.

The first equality expresses *inner regularity* by closed sets, and the second expresses *outer regularity* by open sets. Both follow from (14.1) because $\mu(F) \le \mu(E) \le \mu(U)$ and $\mu(U) - \mu(F) = \mu(U \setminus F) < \varepsilon$, where finiteness of μ allows the subtraction.

On $\mathbb{R}^d$, a measure that is finite on compact sets is finite on every ball, so it can be restricted to a large ball to obtain a finite Borel measure. Combining the two theorems in this way yields density of compactly supported continuous functions. For $g\colon \mathbb{R}^d \to \mathbb{R}$, the **support** $\operatorname{spt} g$ is the closure of $\{g \neq 0\}$, and $C_c(\mathbb{R}^d)$ denotes the continuous functions with compact support. If $\mu(K) < +\infty$ for every compact $K \subseteq \mathbb{R}^d$, then each $g \in C_c(\mathbb{R}^d)$ satisfies $\int |g|^p \, d\mu \le (\max |g|)^p \mu(\operatorname{spt} g) < +\infty$, so $C_c(\mathbb{R}^d) \subseteq L^p(\mathbb{R}^d, \mu)$.

Corollary 14.5 (Density of C_c on $\mathbb{R}^d$). Suppose that μ is a Borel measure on $\mathbb{R}^d$ with $\mu(K) < +\infty$ for every compact set $K \subseteq \mathbb{R}^d$, and $p \in (0, +\infty)$. Then $C_c(\mathbb{R}^d)$ is dense in $L^p(\mathbb{R}^d, \mu)$.

A Borel measure that is finite on compact sets is σ-finite, because $\mathbb{R}^d$ is the union of the closed balls $\overline{B}(0, n)$, $n \in \mathbb{N}$. In particular, Corollary 14.5 applies to Lebesgue measure on $\mathbb{R}^d$. The converse fails, and σ-finiteness alone does not suffice for Corollary 14.5.

Warning (A σ-finite Borel measure without dense continuous functions). On $\mathbb{R}$, the counting measure on the rationals, $\mu(A) \coloneqq \#(A \cap \mathbb{Q})$, is a σ-finite Borel measure: $\mathbb{R}$ is the union of the null set $\mathbb{R} \setminus \mathbb{Q}$ and the countably many singletons $\{q\}$, $q \in \mathbb{Q}$, each of measure 1. If a continuous function g satisfies $g(x_0) \neq 0$, then $|g| > \frac{1}{2}|g(x_0)|$ on an interval around x_0, which contains infinitely many rationals, so $\int |g|^p \, d\mu = +\infty$. Hence the only continuous function in $L^p(\mathbb{R}, \mu)$ is 0, while $\mathbb{1}_{\{0\}}$ is a non-zero element of $L^p(\mathbb{R}, \mu)$. The interval $[0, 1]$ is compact but $\mu([0, 1]) = +\infty$, so the hypothesis of Corollary 14.5 fails.

The density of continuous functions also fails for $p = \infty$, even for Lebesgue measure on $[0, 1]$.

Warning (Continuous functions are not dense in L^∞). On $[0,1]$ with Lebesgue measure, the indicator $f \coloneqq \mathbb{1}_{[0,1/2]}$ satisfies $\|f-g\|_\infty \geq \frac{1}{2}$ for every continuous $g\colon [0,1] \to \mathbb{R}$. Indeed, suppose that $\delta \coloneqq \|f-g\|_\infty < \frac{1}{2}$. Then $g \geq 1-\delta$ almost everywhere on $[0,\frac{1}{2}]$ and $|g| \leq \delta$ almost everywhere on $(\frac{1}{2},1]$. A set of full measure in an interval is dense in that interval, so continuity of g gives $g(\frac{1}{2}) \geq 1-\delta$ and $g(\frac{1}{2}) \leq \delta$. Hence $1-\delta \leq \delta$, which contradicts $\delta < \frac{1}{2}$.

14.3 Proof of regularity

The proof of Theorem 14.4 does not construct F and U for each Borel set directly. Instead, it shows that the collection of sets satisfying (14.1) contains the closed sets and is a σ-algebra; since the Borel σ-algebra is the smallest σ-algebra containing the open sets, and hence the closed sets, the collection contains every Borel set. For closed sets, the open sets U are neighborhoods defined through the distance function.

For a non-empty set $A \subseteq X$ and $x \in X$, the **distance** from x to A is $\operatorname{dist}(x,A) \coloneqq \inf_{a\in A} d(x,a)$.

Lemma 14.6 (Distance to a set). For a non-empty set $A \subseteq X$, the function $x \mapsto \operatorname{dist}(x,A)$ satisfies $|\operatorname{dist}(x,A) - \operatorname{dist}(y,A)| \leq d(x,y)$ for all $x,y \in X$, so it is continuous. Moreover, $\operatorname{dist}(x,A) = 0$ if and only if x belongs to the closure $\overline{A}$.

Proof of Lemma 14.6. For every $a \in A$, the triangle inequality gives $\operatorname{dist}(x,A) \leq d(x,a) \leq d(x,y) + d(y,a)$. Taking the infimum over $a \in A$ yields $\operatorname{dist}(x,A) - \operatorname{dist}(y,A) \leq d(x,y)$, and exchanging x and y gives the first claim. The equality $\operatorname{dist}(x,A) = 0$ holds if and only if there are points $a_n \in A$ with $d(x,a_n) \to 0$, which is the condition $x \in \overline{A}$. □

Proof of Theorem 14.4. Let $\mathcal{G}$ denote the collection of Borel sets $E \subseteq X$ such that for every $\varepsilon > 0$ there exist a closed set F and an open set U satisfying (14.1).

Step 1: Closed sets belong to $\mathcal{G}$.
The empty set belongs to $\mathcal{G}$ with $F = U = \emptyset$. Suppose that $F \neq \emptyset$ is closed, and define $U_n \coloneqq \{x \in X : \operatorname{dist}(x,F) < 1/n\}$ for $n \in \mathbb{N}$. By Lemma 14.6, each U_n is open, $U_n \supseteq F$, the sets U_n decrease, and

$$\bigcap_{n=1}^{+\infty} U_n = \{x \in X : \operatorname{dist}(x,F) = 0\} = \overline{F} = F.$$

Because $\mu(U_1) \leq \mu(X) < +\infty$, continuity of measure from above gives $\mu(U_n \setminus F) \to 0$. For $\varepsilon > 0$, the pair F, U_n with n large satisfies (14.1).

Step 2: Closure under complements.
If $F \subseteq E \subseteq U$ with F closed and U open, then $X \setminus U \subseteq X \setminus E \subseteq X \setminus F$, where $X \setminus U$ is closed and $X \setminus F$ is open, and

$$(X \setminus F) \setminus (X \setminus U) = U \setminus F.$$

Hence $E \in \mathcal{G}$ implies $X \setminus E \in \mathcal{G}$ with the same estimate.

Step 3: Closure under countable unions.
Suppose that $E_k \in \mathcal{G}$ for $k \in \mathbb{N}$, set $E \coloneqq \bigcup_k E_k$, and fix $\varepsilon > 0$. For each k, choose a closed set F_k and an open set U_k with $F_k \subseteq E_k \subseteq U_k$ and $\mu(U_k \setminus F_k) < 2^{-k}\varepsilon$. The set $U \coloneqq \bigcup_k U_k$ is open and contains E. Because $U \setminus \bigcup_k F_k \subseteq \bigcup_k (U_k \setminus F_k)$, countable subadditivity gives

$$\mu\Big(U \setminus \bigcup_{k=1}^{+\infty} F_k\Big) \leq \sum_{k=1}^{+\infty} 2^{-k}\varepsilon = \varepsilon.$$

A countable union of closed sets need not be closed, so we replace $\bigcup_k F_k$ by the finite unions $F^{(N)} \coloneqq \bigcup_{k=1}^{N} F_k$, which are closed and contained in E. The sets $U \setminus F^{(N)}$ decrease to $U \setminus \bigcup_k F_k$ as $N \to \infty$, and $\mu(U) < +\infty$, so continuity of measure from above gives $\mu(U \setminus F^{(N)}) < 2\varepsilon$ for N large. Because $\varepsilon > 0$ was arbitrary, $E \in \mathcal{G}$.

Step 4: Conclusion for closed and open sets.
By Steps 2 and 3, $\mathcal{G}$ is a σ-algebra, and by Step 1 it contains the closed sets, hence also their complements, the open sets. Because $\mathcal{B}(X)$ is the smallest σ-algebra containing the open sets, $\mathcal{B}(X) \subseteq \mathcal{G}$.

Step 5: Compact sets on $\mathbb{R}^d$.
Suppose that $X = \mathbb{R}^d$, E is a Borel set, and $\varepsilon > 0$. By Step 4, there exist a closed set F and an open set U with $F \subseteq E \subseteq U$ and $\mu(U \setminus F) < \varepsilon$. The sets $K_n \coloneqq F \cap \overline{B}(0, n)$ are closed and bounded, hence compact by the Heine–Borel theorem, and they increase to F. The sets $F \setminus K_n$ decrease to $\emptyset$, so continuity of measure from above gives $\mu(F \setminus K_n) < \varepsilon$ for n large. For such n, the compact set $K_n \subseteq E$ satisfies $\mu(U \setminus K_n) = \mu(U \setminus F) + \mu(F \setminus K_n) < 2\varepsilon$. Because $\varepsilon > 0$ was arbitrary, F in (14.1) can be chosen compact. □

14.4 Proof of density

The regularity theorem provides, for a Borel set E, a closed set $F \subseteq E$ and an open set $U \supseteq E$ with $\mu(U \setminus F)$ small. To approximate $\mathbb{1}_E$, we need a continuous function equal to 1 on F and to 0 outside U. Distance functions provide such a function explicitly.

Lemma 14.7 (Urysohn functions on metric spaces). Suppose that $F \subseteq U \subseteq X$ with F closed and U open. Then there exists a continuous function $\varphi \colon X \to [0, 1]$ such that

$$\mathbb{1}_F \leq \varphi \leq \mathbb{1}_U. \tag{14.2}$$

Proof of Lemma 14.7. If $F = \emptyset$, the function $\varphi \coloneqq 0$ satisfies (14.2); if $U = X$, the function $\varphi \coloneqq 1$ does. Otherwise, F and $X \setminus U$ are non-empty, and we define

$$\varphi(x) \coloneqq \frac{\operatorname{dist}(x, X \setminus U)}{\operatorname{dist}(x, X \setminus U) + \operatorname{dist}(x, F)}.$$

The denominator is positive at every $x \in X$: by Lemma 14.6, if both distances vanished, then x would belong to the closed sets $X \setminus U$ and F, which are disjoint because $F \subseteq U$. Hence φ is continuous by Lemma 14.6, and $0 \leq \varphi \leq 1$. For $x \in F$, $\operatorname{dist}(x, F) = 0$, so $\varphi(x) = 1$; for $x \notin U$, $\operatorname{dist}(x, X \setminus U) = 0$, so $\varphi(x) = 0$. □

If $F \subseteq E \subseteq U$ and φ satisfies (14.2), then $\mathbb{1}_E$ and φ both equal 1 on F, both equal 0 outside U, and both take values in $[0, 1]$ on $U \setminus F$. Hence

$$|\mathbb{1}_E - \varphi| \leq \mathbb{1}_{U \setminus F} \qquad \text{and} \qquad \int_X |\mathbb{1}_E - \varphi|^p \, \mathrm{d}\mu \leq \mu(U \setminus F). \tag{14.3}$$

Proof of Theorem 14.3. Fix $f \in L^p(X, \mu)$ and $\varepsilon > 0$. We use the constant $K_p = \max\{1, 2^{p-1}\}$ and the quasi-triangle inequalities for $\|\cdot\|_p^p$ from Note 25.

Step 1: Reduction to a simple function.
By Theorem 14.1, there is a simple function $s = \sum_{i=1}^{m} a_i \mathbb{1}_{E_i}$ with $a_i \neq 0$ and Borel sets E_i such that $\|f - s\|_p^p < \varepsilon$.

Step 2: Approximation of each indicator.
Because s is now fixed, we may choose $\eta > 0$ such that $\eta \sum_{j=1}^{m} K_p^j |a_j|^p < \varepsilon$. For each i, Theorem 14.4 provides a closed set F_i and an open set U_i with $F_i \subseteq E_i \subseteq U_i$ and $\mu(U_i \setminus F_i) < \eta$. Lemma 14.7 provides $\varphi_i \in C_b(X)$ with $\mathbb{1}_{F_i} \leq \varphi_i \leq \mathbb{1}_{U_i}$, and (14.3) gives $\|\mathbb{1}_{E_i} - \varphi_i\|_p^p \leq \mu(U_i \setminus F_i)$.

Step 3: Approximation of the simple function.
The function $g := \sum_{i=1}^{m} a_i \varphi_i$ belongs to $C_b(X)$. At every point, $|s - g| \leq \sum_{i=1}^{m} |a_i| \, |\mathbb{1}_{E_i} - \varphi_i|$, and the iterated quasi-triangle inequality (Note 25) gives

$$|s - g|^p \leq \sum_{i=1}^{m} K_p^i |a_i|^p \, |\mathbb{1}_{E_i} - \varphi_i|^p.$$

Integrating and using Step 2 yields

$$\|s - g\|_p^p \leq \sum_{i=1}^{m} K_p^i |a_i|^p \, \mu(U_i \setminus F_i) < \eta \sum_{i=1}^{m} K_p^i |a_i|^p < \varepsilon.$$

Step 4: Conclusion.
The quasi-triangle inequality for $\|\cdot\|_p^p$ gives

$$\|f - g\|_p^p \leq K_p(\|f - s\|_p^p + \|s - g\|_p^p) < 2K_p\varepsilon.$$

Because $\varepsilon > 0$ was arbitrary, $C_b(X)$ is dense in $L^p(X, \mu)$. □

The corollary on $\mathbb{R}^d$ follows the same pattern, with two additional reductions: a truncation to a ball, which makes the measure finite, and the compact sets from Theorem 14.4, which make the support of the approximating function compact. The details are left to Homework 08.